

YEAR 7 • UNIT 1 • HOMEWORK 3

Factors, Tests and Tissues

Mathematics 1.4 — Highest Common Factors • 1.5 — Tests for Divisibility

Science 1.4 — Cells, Tissues and Organs

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the division, not just the answer**. Almost every question in Section A and Section B is really a division question.
- In science, answer in **full English sentences**. “Organ” is not a sentence; “The heart is an organ” is. If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A • Mathematics 1.4 — Highest Common Factors

Remember from the lesson

A **factor** of a number divides into it **exactly**, with nothing left over. You find factors by hunting for **pairs**: $24 = 1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6$.

Common means **shared**. A **common factor** is in both lists, and the **highest common factor (HCF)** is the biggest of them — the same idea you already know as **UCLN**.

The HCF is never “none”. 1 is a factor of every number, so every pair has at least one common factor.

A1 Hunt the pairs

(a) Write **every** pair of whole numbers that multiplies to **36**, then list all the factors of 36.

Pairs: _____

Factors of 36: _____

(b) Now do the same for **54**.

Pairs: _____

Factors of 54: _____

(c) Which numbers are in **both** lists? _____

(d) So the **HCF** of 36 and 54 is _____

A2 Divide to decide

The only way to be sure

There is one test that always works: **do the division**. If there is **nothing left over**, it is a factor. If there is a **remainder**, it is not.

Do each division. Write the answer, and the remainder if there is one. Then tick the box only if the first number is a **factor**.

Division	Answer	Factor?	Division	Answer	Factor?
$96 \div 6$		☐	$96 \div 7$		☐
$84 \div 4$		☐	$84 \div 5$		☐
$105 \div 7$		☐	$132 \div 8$		☐
$156 \div 12$		☐	$210 \div 6$		☐

A3 Say it, then write it

Word help – the English tells you which number to start with

divide 96 by 8

you **start** at 96: $96 \div 8$.

share 72 equally between 9

divide: $72 \div 9$.

6 divides into 84

$84 \div 6$, not $6 \div 84$.

exactly

with **nothing left over**. No remainder.

the highest common factor of

the **biggest** number that divides into **both**.

in its simplest form

written with the **smallest** numbers you can.

Turn each sentence into a calculation, then work it out. **Write the calculation first.**

1. Divide 96 by 8.

Calculation: _____ Answer: _____

2. Share 72 equally between 9.

Calculation: _____ Answer: _____

3. Find the highest common factor of 20 and 30.

Calculation: _____ Answer: _____

4. Does 6 divide into 81 exactly? Show the division.

Calculation: _____ Answer: _____

A4 What the HCF is actually for**Simplest form in one step**

Find the HCF of the top and the bottom, then **divide both of them by it**. That is the whole method – and it is two more divisions.

Write each fraction in its simplest form. Fill in every box.

Fraction	HCF	Top $\div$ HCF	Bottom $\div$ HCF	Simplest form
$\frac{18}{24}$				
$\frac{30}{45}$				
$\frac{48}{60}$				

A5 Two answers that surprise people**Work these out before you read on**

Both of these pairs catch nearly everybody. Do the factor lists if you need to.

(a) The HCF of **9** and **10** is _____

(b) The HCF of **7** and **42** is _____

(c) A student writes that 9 and 10 have **no** common factor. In one full sentence, explain why that can never be true for any pair of numbers.

A6 Word problems

Read each one **all the way to the end** before you start writing.

1. **The bracelets.** Mr Bowen has **42 red beads** and **56 blue beads**. He makes identical bracelets from them, using every bead, with **nothing left over**, and he wants **as many bracelets as possible**. How many bracelets can he make, and how many beads of each colour go on one bracelet?

2. **The test.** There are **60 questions** on a test. Mr Bowen gets **45** of them right. Write the

fraction he got right, in its **simplest form**.

Section B · Mathematics 1.5 – Tests for Divisibility

The nine tests you wrote down

- by 2** the last digit is 0, 2, 4, 6 or 8
- by 3** the digits add up to a multiple of 3
- by 4** the last **two** digits make a multiple of 4
- by 5** the last digit is 0 or 5
- by 6** it passes the test for 2 **and** the test for 3
- by 8** the last **three** digits make a multiple of 8
- by 9** the digits add up to a multiple of 9
- by 10** the last digit is 0
- by 11** split the digits into two groups, taking every other one, and add each group.
The **difference** is 0 or a multiple of 11

There is **no easy test for 7**. For 7 you divide and look at the remainder.

B1 The factor hunt

Use the tests. Put a **tick** in the box if the number on the left divides by the number at the top, and a **cross** if it does not. Do **not** use a calculator.

	2	3	4	5	6	8	9	10	11
1260									
3465									
2088									
4510									
5832									

Now pick **one** of your ticks and say which test proved it.

I know that _____ divides by _____ because _____

B2 Now prove it

A test says yes or no. Division says how many.

A divisibility test is a **shortcut**: it tells you there is nothing left over, but it never tells you the **answer**. For that you still have to divide.

Each of these came out as a tick in B1. Carry out the division and write the answer.

(a) $1260 \div 4 =$ _____

(d) $2088 \div 8 =$ _____

(b) $1260 \div 9 =$ _____

(e) $4510 \div 5 =$ _____

(c) $3465 \div 11 =$ _____

(f) $5832 \div 6 =$ _____

(g) Two of those six have the **same** answer. Which two? _____

B3 The missing digit

In each one a digit has been rubbed out. Use a test to work out what it must be. Show the digit sum or the groups you used.

1. $52\Box$ is divisible by 3. There is more than one answer — find them all.

Working: _____ Digit(s): _____

2. $741\Box$ is divisible by 9.

Working: _____ Digit: _____

3. $\Box630$ is divisible by 11.

Working: _____ Digit: _____

B4 Word problems

1. **The stickers.** Mr Bowen has **4176 stickers** to share equally between the **8** classes in the school. Use a test to decide whether he can do it with none left over — then work out how many each class gets.

2. **The socks.** Mr Bowen has **2464 socks**. He puts them into **pairs**, with none left over, and then puts the pairs into drawers of **11 pairs** each, again with none left over. Show that he can do both, and work out how many drawers he needs.

B5 Now you write the question**Your turn to be the teacher**

So far the questions have given you the English and you have written the maths. Now do it **backwards**.

Your word problem must:

- be about something **real**, written in **full English sentences**, and end with a question;
- share things out **equally with nothing left over**;
- have the answer that is already given to you.

(a) Write a word problem for

$$4368 \div 8 = 546$$

(b) How could you tell that 4368 divides by 8 **without** doing the division? Answer in one full sentence.

Section C · Science 1.4 – Cells, Tissues and Organs**Remember from the lesson**

The word that does all the work in this section is **similar** against **different**. **Similar** cells make a **tissue**. **Different** tissues make an **organ**.

C1 Match the word to its meaning

Write the correct letter in the box next to each word.

Word	Letter	Meaning
1. Tissue		A A living thing, which may contain many organ systems.
2. Organ		B A group of similar cells, all doing one particular job.
3. Organ system		C A structure made of several different tissues working together.
4. Organism		D A set of organs that all work together to do the same job.
5. Epithelium		E The tall tissue near the top of a leaf, packed with chloroplasts.
6. Palisade layer		F A tissue that covers a surface.

C2 Which level is it?

Write **cell**, **tissue**, **organ**, **organ system** or **organism** for each one.

What it is	Which level?	What it is	Which level?
a red blood cell		the digestive system	
a leaf		onion epidermis	
the palisade layer		Mr Bowen	
the heart		a root hair cell	

Two of those eight catch almost everybody: **a leaf** and **the palisade layer**. In one full sentence each, say how you decided.

A leaf is _____

The palisade layer is _____

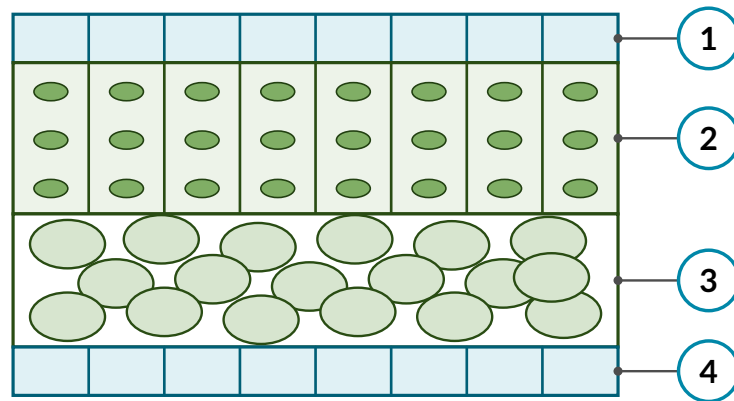
C3 Climb the ladder

One example, climbed all five rungs. The first is done for you.

Level	The example
Cell	a ciliated cell
Tissue	
Organ	
Organ system	
Organism	

C4 Label the leaf

A leaf is one **organ** made of **four different tissues**, cut open here. Name each tissue and say what it does.



No.	Name of the tissue	Its job
1		
2		
3		
4		

C5 Answer in full sentences

Word help – one word, two completely different meanings

a tissue (everyday) a paper handkerchief. You can count them: *a tissue, two tissues.*
tissue (science) a group of similar cells. You do **not** count it: *muscle tissue, not a muscle tissue.*

- Write **two sentences of your own**. In the first, use *tissue* with its everyday meaning. In the second, use *tissue* with its scientific meaning.

- A jellyfish has no heart, no lungs and no brain. Which level of the ladder can a living thing manage without? Explain your answer.

Answer Key – Teacher Copy

A1 Hunt the pairs

(a) $36 = 1 \times 36, 2 \times 18, 3 \times 12, 4 \times 9, 6 \times 6$. Factors: 1, 2, 3, 4, 6, 9, 12, 18, 36.

(b) $54 = 1 \times 54, 2 \times 27, 3 \times 18, 6 \times 9$. Factors: 1, 2, 3, 6, 9, 18, 27, 54.

(c) 1, 2, 3, 6, 9, 18

(d) 18

6×6 is the one people miss, because it is the same number twice – and it is why 36 has an odd number of factors. Worth a sentence at the board if several students list eight factors instead of nine.

A2 Divide to decide

$96 \div 6 = 16$ **yes** $96 \div 7 = 13 \text{ r } 5$ **no** $84 \div 4 = 21$ **yes** $84 \div 5 = 16 \text{ r } 4$ **no**

$105 \div 7 = 15$ **yes** $132 \div 8 = 16 \text{ r } 4$ **no** $156 \div 12 = 13$ **yes** $210 \div 6 = 35$ **yes**

Five ticks and three crosses. The one to watch is $132 \div 8$: a student who has learned “even numbers divide by 8” ticks it without dividing. Make them show the remainder. $156 \div 12$ is the hardest division on the page – if it is the only one they get wrong, that is an arithmetic problem, not a factors problem.

A3 Say it, then write it

1. $96 \div 8 = 12$

2. $72 \div 9 = 8$

3. $\text{HCF}(20, 30) = 10$. Factors of 20: 1, 2, 4, 5, 10, 20. Of 30: 1, 2, 3, 5, 6, 10, 15, 30. Common: 1, 2, 5, 10.

4. $81 \div 6 = 13 \text{ r } 3$, so **no** – 6 does not divide into 81 exactly.

Question 4 is the one that tests whether “exactly” has landed. An answer of “13.5” is not wrong arithmetic but it is the wrong question: at this stage we want the remainder, because the remainder is what decides whether it is a factor.

A4 What the HCF is actually for

18/24 HCF 6 $18 \div 6 = 3$ $24 \div 6 = 4$ **3/4**

30/45 HCF 15 $30 \div 15 = 2$ $45 \div 15 = 3$ **2/3**

48/60 HCF 12 $48 \div 12 = 4$ $60 \div 12 = 5$ **4/5**

The predictable error is using *a* common factor rather than the *highest* one – dividing 18/24 by 2 to get 9/12 and stopping. That is not wrong, it is unfinished, and the fix is to say so in those words. Six divisions live in this question; if a student’s HCF is right and the fraction is wrong, the failure is division, not factors.

A5 Two answers that surprise people

(a) **1** – not “none”. 9 and 10 are consecutive, so 1 is the only number in both lists.

(b) **7** – not 1. 7 divides into 42, so the HCF is just the smaller number.

(c) Any sentence saying that **1 is a factor of every number**, so every pair of numbers has at least one common factor.

These are the two mirror-image errors from the lesson, and they are on the page together on purpose. Expect “none” for (a) and “1” for (b) from the same students. Mark (c) strictly: it must name 1, not just say “it is never true”.

A6 Word problems

1. $\text{HCF}(42, 56) = 14$, so **14 bracelets**, each with $42 \div 14 = 3$ red beads and $56 \div 14 = 4$ blue beads. (Factors of 42: 1, 2, 3, 6, 7, 14, 21, 42. Of 56: 1, 2, 4, 7, 8, 14, 28, 56.)
2. $45/60$. $\text{HCF}(45, 60) = 15$, so $45 \div 15 = 3$ and $60 \div 15 = 4$: **3/4**.

In problem 1, students who answer “14” have done the maths but not the question — the beads per bracelet are two more divisions and they are what the HCF was for. Problem 2 is the same method as A4 wearing a story; if a student can do A4 and not this, the gap is reading, not arithmetic.

B1 The factor hunt

1260 yes: 2, 3, 4, 5, 6, 9, 10 no: 8, 11

3465 yes: 3, 5, 9, 11 no: 2, 4, 6, 8, 10

2088 yes: 2, 3, 4, 6, 8, 9 no: 5, 10, 11

4510 yes: 2, 5, 10, 11 no: 3, 4, 6, 8, 9

5832 yes: 2, 3, 4, 6, 8, 9 no: 5, 10, 11

Digit sums: 1260 gives 9, 3465 gives 18, 2088 gives 18, 4510 gives 10, 5832 gives 18. Last three digits for the test for 8: 260 (no), 465 (no), 088 (yes, $88 \div 8 = 11$), 510 (no), 832 (yes, $832 \div 8 = 104$). Groups for the test for 11: 1260 gives 7 and 2 (difference 5, no); 3465 gives 9 and 9 (difference 0, **yes**); 2088 gives 10 and 8 (difference 2, no); 4510 gives 5 and 5 (difference 0, **yes**); 5832 gives 8 and 10 (difference 2, no).

The row worth marking carefully is **1260**: it divides by 4 but *not* by 8, which kills “divisible by 4 means divisible by 8”. The other trap is **4510** — it is even, so students tick 6 without checking the digit sum, which is 10 and fails the test for 3. Both errors are conceptual, not careless.

B2 Now prove it

(a) 315 (b) 140 (c) 315 (d) 261 (e) 902 (f) 972

(g) **(a) and (c)** — both come to 315.

This question is the division practice of the packet and it is where weak arithmetic will show. (c) $3465 \div 11$ is the hardest; (e) $4510 \div 5$ is the easiest and is there so everybody finishes something. Part (g) is not decoration: it makes them look back at their own six answers, which is the check they never do voluntarily.

B3 The missing digit

1. $5 + 2 = 7$. The digit sum must reach 9, 12 or 15, so the digit is **2, 5 or 8** (giving 522, 525, 528).
2. $7 + 4 + 1 = 12$. The next multiple of 9 is 18, and $18 - 12 = 6$, so the digit is **6** and the number is 7416. (Only one answer: the multiple after 18 is 27, which would need a digit of 15.)
3. Groups are $\square + 3$ and $6 + 0 = 6$. For a difference of 0 we need $\square + 3 = 6$, so the digit is **3** and the number is 3630. ($3630 \div 11 = 330$.)

Question 1 is the one that tests whether they believe an answer can be a set rather than a number — most will find one digit and stop. Ask “are you sure that is the only one?” rather than marking it wrong.

B4 Word problems

1. The last three digits are 368, and $368 \div 8 = 46$, so 4176 divides by 8: **yes**, with none left over. $4176 \div 8 = 522$ stickers each.
2. 2464 is even, so it makes $2464 \div 2 = 1232$ pairs. For 11: the groups are $2 + 6 = 8$ and $4 + 4 = 8$, a difference of 0, so 2464 divides by 11 — and so does 1232, giving $1232 \div 11 = 112$ drawers. Mr Bowen has two feet.

Problem 2 is the deadpan one. Read it completely straight and do not explain the joke; the maths is correct and that is the whole point. It is also a two-step division, which is harder than it looks — a student who divides 2464 by 11 directly gets 224, which is the number of socks per drawer, not the number of drawers. Both routes are defensible if they say which they took.

B5 Now you write the question

No single right answer. Check, in this order:

1. **Is it a sharing story?** Something split into 8 equal groups, or packed 8 to a box. A story where 8 things are bought is a multiplication, not this.
2. **Does it say “equally” or “with nothing left over”?** Without that, the question does not have a single answer.
3. **Full sentences, ending in a question?**
4. **Does their own answer come out as 546?**

(b) The last three digits are **368**, and $368 \div 8 = 46$, so 4368 divides by 8.

One that works for (a): “A factory packs 4368 eggs into boxes. Each box holds 8 eggs, and every box is full. How many boxes do they fill?”

Part (b) is the one worth marking hardest: it is the difference between knowing the test and having memorised an answer. A student who writes “because I divided it” has answered the opposite of the question that was asked.

C1 Match the word to its meaning

1-B 2-C 3-D 4-A 5-F 6-E

C2 Which level is it?

a red blood cell	cell	the digestive system	organ system
a leaf	ORGAN	onion epidermis	tissue
the palisade layer	TISSUE	Mr Bowen	organism
the heart	organ	a root hair cell	cell

A leaf is an **organ** because it is made of four *different* tissues. The palisade layer is a **tissue** because it is made of cells that are all *similar* and all do one job.

These two are the deliberate traps and they fail in opposite directions: a leaf looks like one flat thing so it gets called a tissue, and the palisade layer has an important-sounding name so it gets called an organ. If the class splits on these two, do not mark it — put the leaf cross-section back on the board and count the four layers out loud.

C3 Climb the ladder

Cell: a ciliated cell → Tissue: ciliated epithelium → Organ: a lung → Organ system: the breathing system → Organism: a human (accept “me”, “Mr Bowen”, “a person”).

The rung most often wrong is the organ: students write “the lungs” or jump straight to “the breathing system”. One lung is the organ; the pair plus the windpipe and the nose is the system. Accept “a lung” or “the lungs” but not “breathing”.

C4 Label the leaf

- | | | |
|---|-----------------|--|
| 1 | upper epidermis | a flat protective skin over the top of the leaf |
| 2 | palisade layer | tall cells packed with chloroplasts, where most photosynthesis happens |
| 3 | spongy layer | loose round cells with air gaps, letting gases move through the leaf |
| 4 | lower epidermis | the protective skin underneath |

1 and 4 are the same tissue in two places, so accept “epidermis” for both only if the student writes *upper* and *lower*. The layer most often mislabelled is 3: it gets called “the air layer” or “the gaps”. The gaps are the point of it, but the tissue is the cells around them.

Coming back to C2: this diagram is the answer to why a leaf is an organ — four different tissues, one structure.

C5 Answer in full sentences

- Everyday: “Mr Bowen sneezed, so he took a tissue out of the box.” Scientific: “The wall of the stomach contains muscle tissue.” Mark the **sentence** — capital letter, full stop, and the countable versus uncountable difference actually showing.
- The **organ system** level, and arguably the organ level too. A jellyfish has cells and tissues but no heart, lungs or brain, so a living thing can exist without organ systems. The bottom two rungs are the ones every organism has.

Question 1 is the real English work in this section and is worth more of your marking time than the rest of C put together. The commonest failure is two sentences that both use the everyday meaning. Question 2 has no tidy answer and should be discussed rather than ticked — accept anything that spots a living thing can manage without the top rungs.